



Hayakawa Electronics (Phils.) Corp

HEPC - NETPULSE

(Network Monitoring System)

USER MANUAL

2026



System Overview

HEPC - NetPulse is a dedicated network monitoring system that provides real-time visibility into tracking latency (ms) and connectivity status for different devices connected to the company's server network.

Installation and Deployment

This system is designed for a plug-and-play deployment:

Deployment Steps:

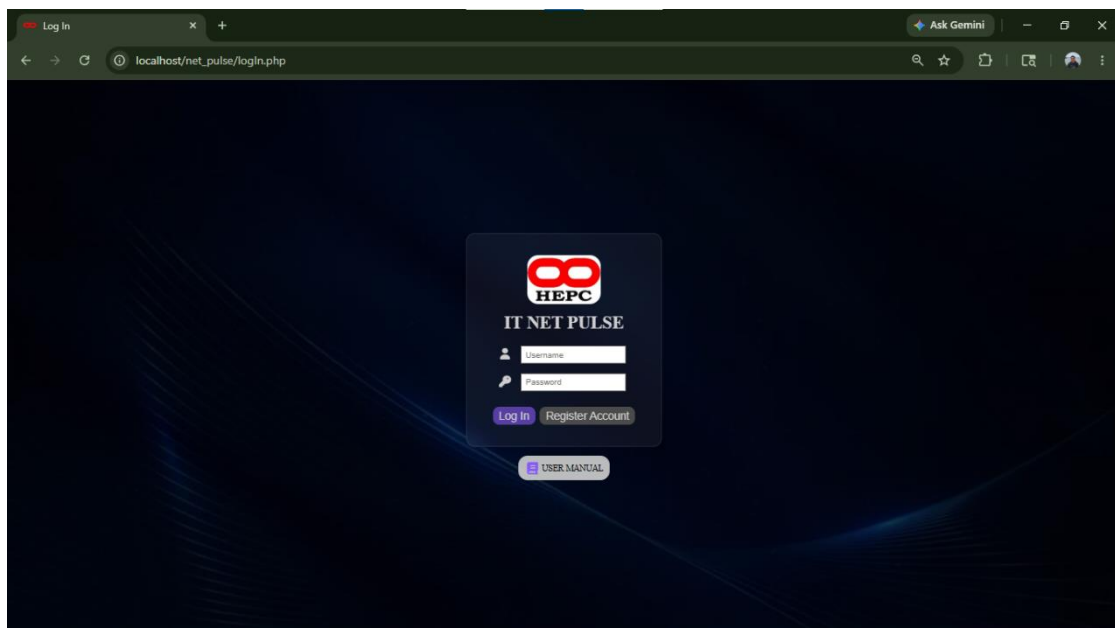
1. Server Requirements: Ensure the target server has XAMPP (or an Apache and MySQL stack) installed.
2. File Placement: Copy the project folder into the htdocs directory of your XAMPP installation folder.
3. Database Setup: Database will be automatically set and initialized on the first load of system login page (e.g., localhost/net_pulse/ login.php or <server-ip>/ net_pulse /login.php).



Use of System

Login Page:

This is the system's landing page. The user will have two options: login or register a new account.



Account Default Credentials:

- Username: admin
- Password: adminpassword

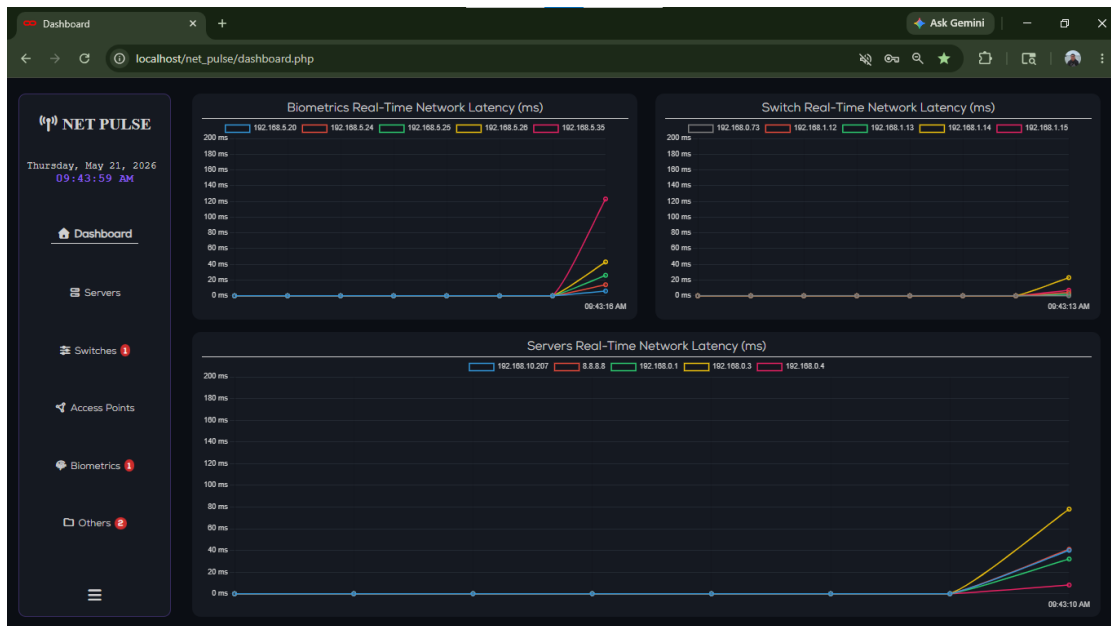
User just need to fill the input fields and click either 'Log In' to log into the system using existing account, or 'Register Account' to register a new account using the data inputted in the fields.



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Dashboard:

The system dashboard displays three (3) line graphs that shows latency (in ms) of the first five (5) devices of three (3) dedicated categories: Servers, Switches, and Biometrics.



Each graph will include the last eight (8) recorded entries for each device of each group.



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Dedicated Device Groups:

NetPulse has four (4) tabs for dedicated device groups: Servers, Switches, Access Points, and Biometrics.

The screenshot shows the 'Servers' tab in the NetPulse application. The left sidebar contains navigation links: Dashboard, Servers (active), Switches, Access Points, Biometrics, and Others. The main content area displays a grid of server status cards. Each card shows a server name, IP address, and a response time in milliseconds (ms). A search bar is located at the top right of the grid. A legend at the bottom indicates signal quality: Green for Excellent Signal (1ms - 89ms), Yellow for Good Signal (90ms - 299ms), Red for Poor Signal (300ms or greater), and Grey for Timed Out / Error.

Server Name	IP Address	Response Time (ms)
SYSTEM SERVER	192.168.10.207	47ms
GOOGLE	8.8.8.8	45ms
192.168.0.1	192.168.0.1	24ms
NOAH	192.168.0.3	60ms
192.168.0.4	192.168.0.4	16ms
192.168.0.7	192.168.0.7	15ms
192.168.0.8	192.168.0.8	5ms
NOAH HRMS	192.168.0.17	3ms
192.168.0.35	192.168.0.35	8ms
192.168.0.36	192.168.0.36	8ms
192.168.0.37	192.168.0.37	3ms
VSPHERE	192.168.0.42	1ms

The screenshot shows the 'Switches' tab in the NetPulse application. The left sidebar contains navigation links: Dashboard, Servers, Switches (active), Access Points, Biometrics, and Others. The main content area displays a grid of switch status cards. Each card shows a switch name, IP address, and a response time in milliseconds (ms). A search bar is located at the top right of the grid. A legend at the bottom indicates signal quality: Green for Excellent Signal (1ms - 89ms), Yellow for Good Signal (90ms - 299ms), Red for Poor Signal (300ms or greater), and Grey for Timed Out / Error.

Switch Name	IP Address	Response Time (ms)
Switch_(IDF_HCT1)	192.168.0.73	--
MGMT_DSW1_IDF1_(IDF01)	192.168.1.12	132ms
MGMT_DSW1_IDF2_(IDF02)	192.168.1.13	3ms
MGMT_DSW1_IDF3_(IDF03)	192.168.1.14	--
MGMT-DSW1-MDF_(MDF)	192.168.1.15	--
MGMT-DSW1-HC01_(Hitachi)	192.168.1.16	--
MGMT_SW01_MDF4_(MDF)	192.168.1.17	--
MGMT_SW02_IDF3_(IDF4)	192.168.1.18	--
MGMT_SW03_IDF3_(IDF4)	192.168.1.19	--
MGMT_SW01_IDF3_(IDF05)	192.168.1.20	--
MGMT_SW04_IDF3_(IDF06)	192.168.1.21	--
MGMT_SW05_IDF3_(IDF08)	192.168.1.22	--
MGMT_SW06_IDF3_(IDF08)	192.168.1.23	--
MGMT_SW08_IDF3_(IDF07)	192.168.1.26	--
MGMT_SW09_IDF3_(IDF13)	192.168.1.31	--



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Access Points

NET PULSE

Thursday, May 21, 2026 09:50:27 AM

Dashboard Servers Switches 1 Access Points Biometrics 1 Others 3

Access Points

HEPC-OF-001 192.168.1.3 (11ms)	HEPC-PR-005 192.168.1.29 (10ms)	HEPC-WH-001 192.168.1.8 (--)	HEPC-PR-004 192.168.1.11 (--)
HEPC-WH-004 192.168.1.30 (--)	HEPC-WH-003 192.168.1.10 (--)	HEPC-GUEST 192.168.1.2 (--)	HEPC-OF-003 192.168.1.25 (--)
HEPC-PR-002 192.168.1.6 (--)	HEPC-PR-003 192.168.1.7 (--)	HEPC-WH-002 192.168.1.9 (--)	HEPC-IT-001 192.168.1.28 (--)
HEPC-OF-002 192.168.1.4 (--)	HEPC-PR-001 192.168.1.5 (--)		

Excellent Signal (1ms - 89ms) Good Signal (90ms - 299ms) Poor Signal (300ms or greater) Timed Out / Error

Biometrics

NET PULSE

Thursday, May 21, 2026 09:50:38 AM

Dashboard Servers Switches 1 Access Points Biometrics 1 Others 3

Biometrics

HEPC IN 192.168.5.20 (105ms)	HEPC OUT 192.168.5.24 (109ms)	HR TEAM OUT 192.168.5.25 (77ms)	HR TEAM IN 192.168.5.26 (87ms)
OFFICE IN & OUT 192.168.5.35 (72ms)	HERU IN 192.168.5.36 (64ms)	HERU OUT 192.168.5.37 (--)	NATCORP IN 192.168.5.38 (--)
NATCORP OUT 192.168.5.39 (--)	HEPC IN 192.168.5.51 (--)	HEPC OUT 192.168.5.52 (--)	HEPC OUT 192.168.5.53 (--)
HERU PROD IN 192.168.5.55 (--)	HERU PROD OUT 192.168.5.57 (--)	HCTI IN 192.168.8.215 (--)	HCTI OUT 192.168.8.216 (--)

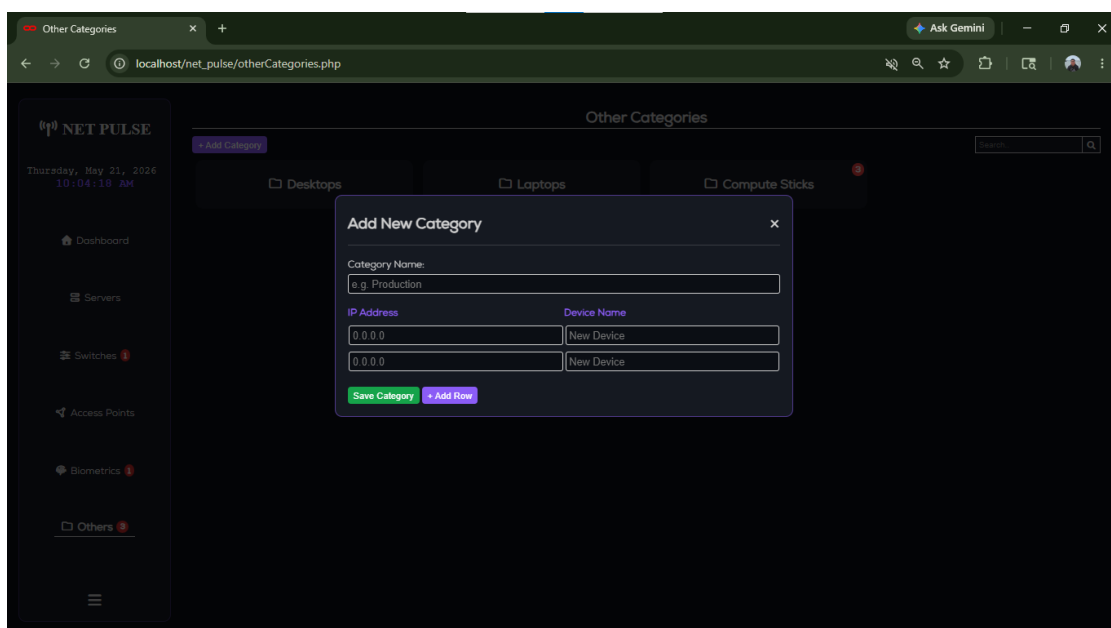
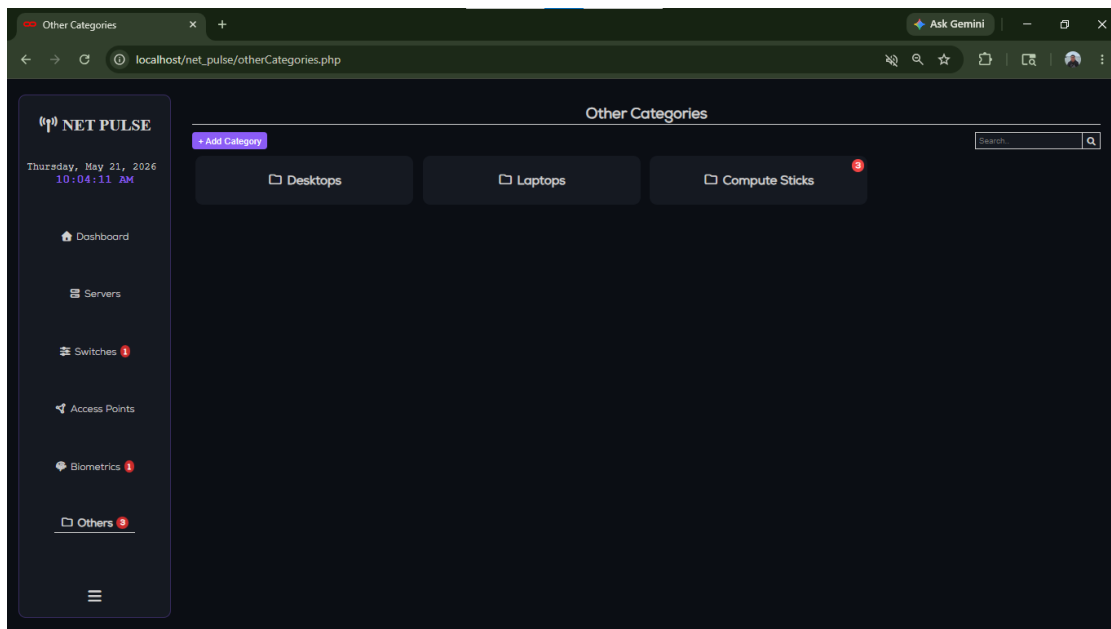
Excellent Signal (1ms - 89ms) Good Signal (90ms - 299ms) Poor Signal (300ms or greater) Timed Out / Error



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Others Category:

This tab's the container of sub-categories of other devices: Desktops, Laptops, Compute Sticks. This section also has the function to add a new category and add addresses for that category.

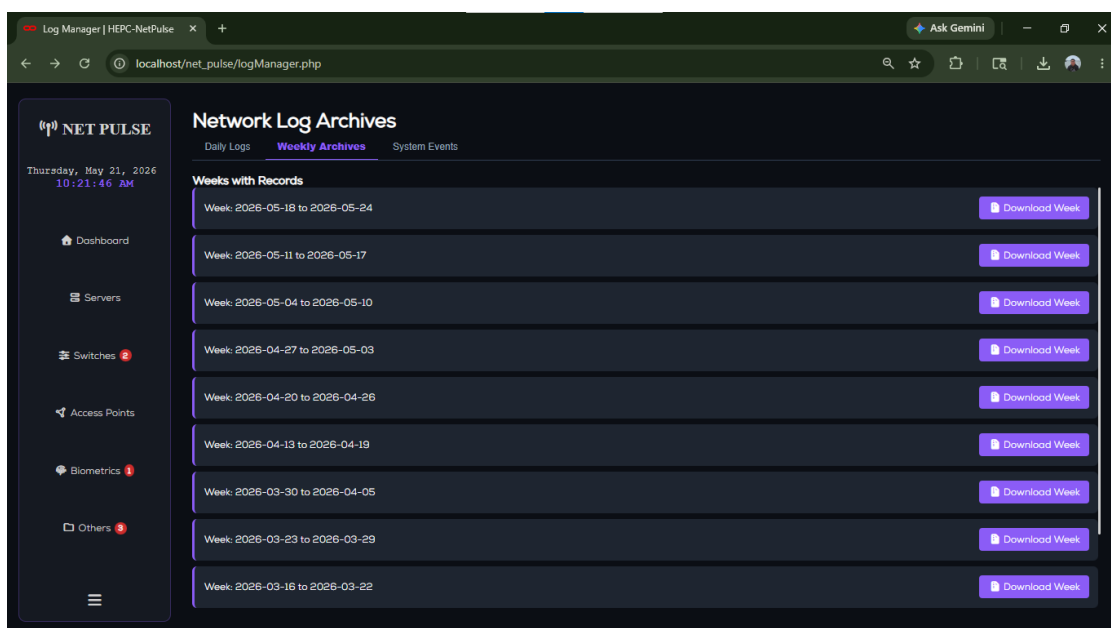
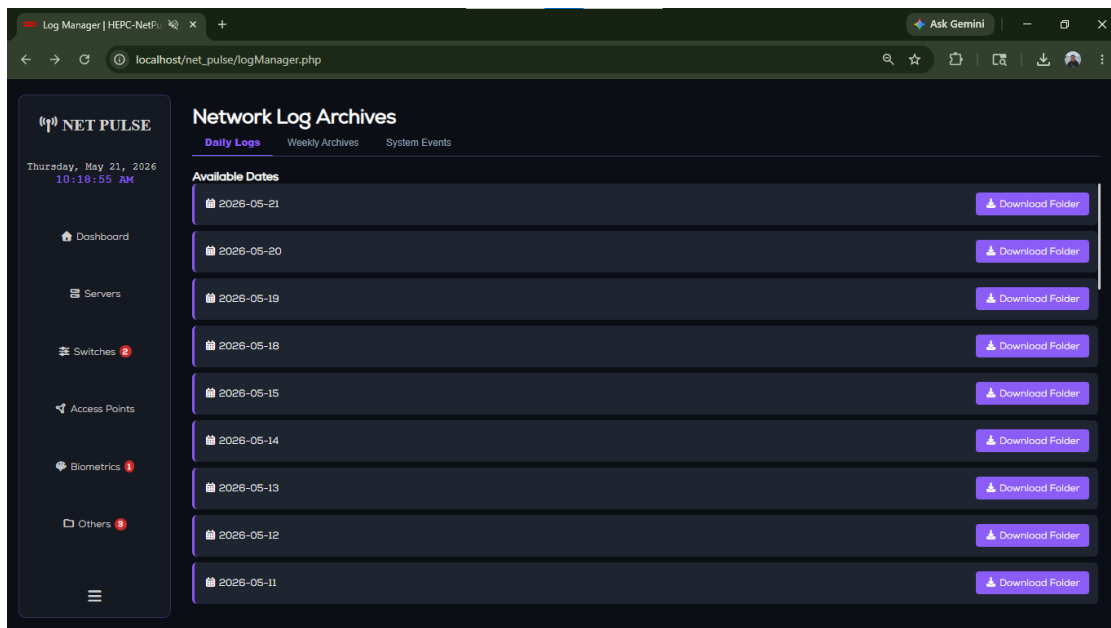




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Logs Handler and Downloader:

This section offers options to download logs for device latencies (which can be dailies or the weekly logs) and system events.





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Log Manager | HEPC-NetPulse x + Ask Gemini

localhost/net_pulse/logManager.php

NET PULSE

Thursday, May 21, 2026
10:22:42 AM

- Dashboard
- Servers
- Switches 2
- Access Points
- Biometrics 1
- Others 1

Network Log Archives

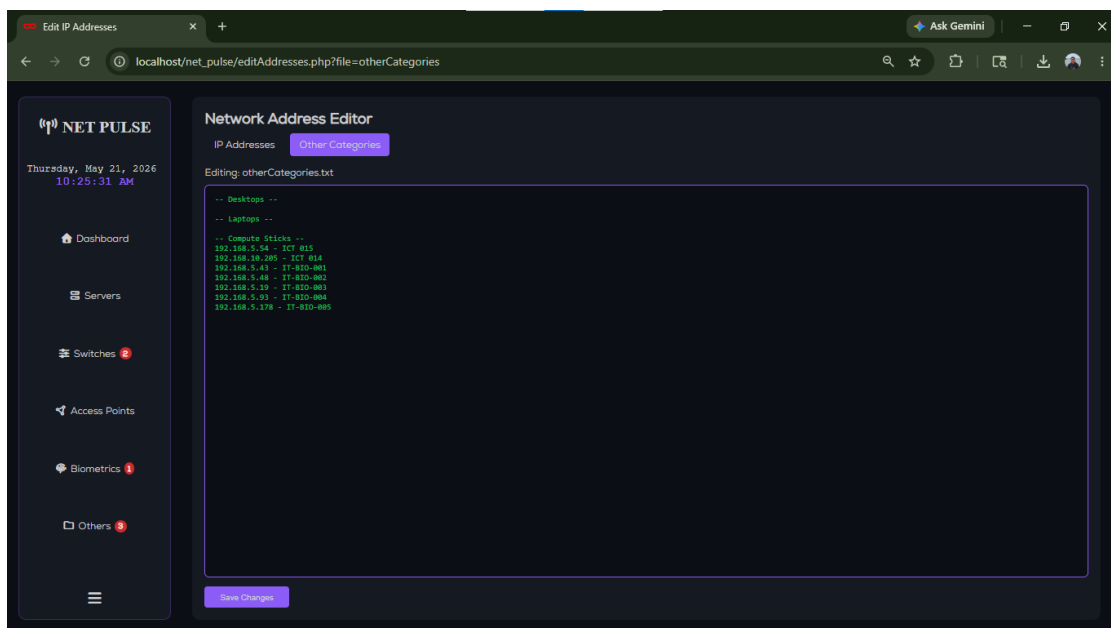
Daily Logs Weekly Archives **System Events**

System Event Logs

2026-05-21-system-events.txt	Download TXT
2026-05-15-system-events.txt	Download TXT
2026-05-14-system-events.txt	Download TXT
2026-05-07-system-events.txt	Download TXT
2026-04-30-system-events.txt	Download TXT
2026-04-24-system-events.txt	Download TXT
2026-04-22-system-events.txt	Download TXT
2026-04-15-system-events.txt	Download TXT
2026-04-14-system-events.txt	Download TXT



NetPulse utilizes two different text files for saving device lists: one for the dedicated categories and another for the others. This function enables the user to edit those text files directly.





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Edit Account Credentials:

This tab displays the credential form, where the user can edit or update their account credentials.

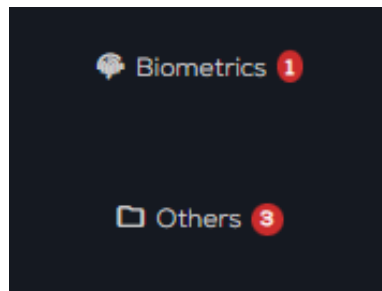
The screenshot shows a web browser window with the address bar displaying 'localhost/net_pulse/editCredentials.php'. The page title is 'Account Credentials'. On the left, there is a sidebar with the 'NET PULSE' logo and a navigation menu including 'Dashboard', 'Servers', 'Switches' (with a red notification badge), 'Access Points', 'Biometrics' (with a red notification badge), and 'Others' (with a red notification badge). The main content area contains a form with the following fields: 'Username' (pre-filled with 'admin'), 'Current Password', 'New Password', and 'Confirm New Password'. A green 'Save' button is located at the bottom right of the form.



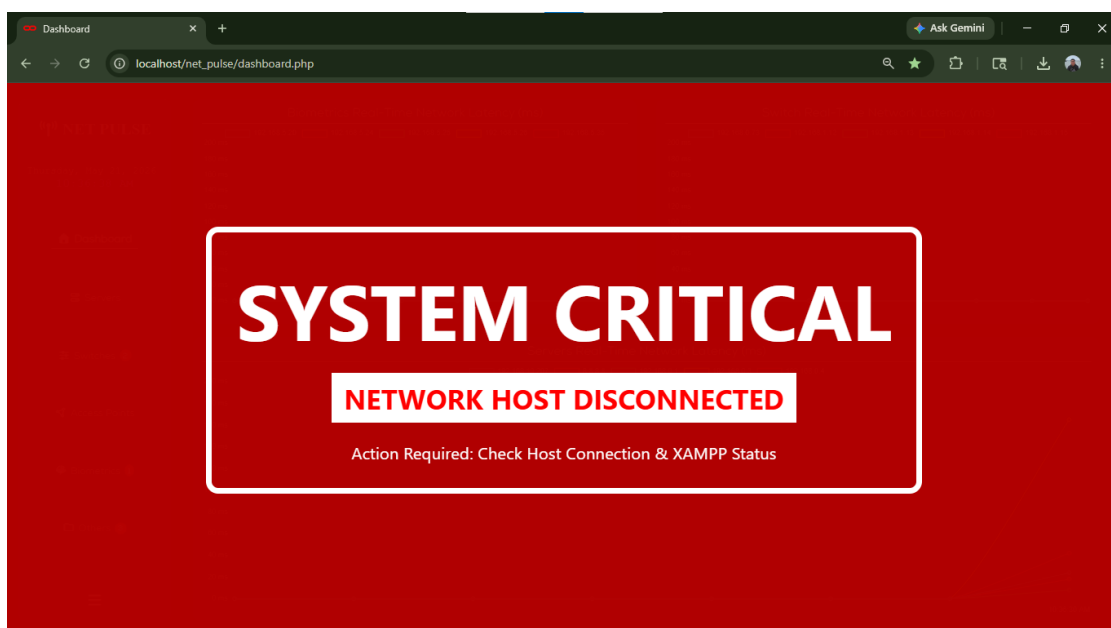
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NetPulse Notification System:

NetPulse utilizes both visual and sound queues to notify the user of offline or downed devices. Each section shows number badges that corresponds number of offline devices.



A quick sound notification will also play for dedicated sections if a device failed to send back a response to the system. The system also has a self-monitoring notification overlay that will pop up if the server goes offline or failed to connect to the local network mesh.





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If the Alert Triggers:

1. Check XAMPP Services: Ensure both Apache and MySQL are running.
2. Verify exec() Permissions: The backend uses the PHP exec() command to run system pings. Ensure this is not disabled in your php.ini file.
3. Network Scope: Ensure the server hosting the app is on a network segment that can "reach" the target IPs (Servers, Switches, Biometrics).
4. Host Network Connectivity: Check if host device is connected to a valid network connection



Security & Stability

- Auto-Logout: A "Heartbeat" script (userHeartbeat.js) monitors admin activity. If the browser tab is closed and inactive for 15 seconds, the session is destroyed and user is automatically logged out for security.
- System Watchdog: A background script (systemWatchdog.js) ensures the application logic remains active and alerts the IT admin if the monitoring process stalls.
- php.ini: Ensure pdo, pdo_mysql, json, session, and zip extensions are enabled in server's php configuration file.

Note: When migrating to a permanent company server, ensure the folder permissions allow the web server (Apache) to write to the local text files and the database directory.